

PiperABM: A Python Library for Resilience-Based Agent Modeling

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Summary

PiperABM is an open-source Python library for building agent-based simulations that couple human behavior with degrading infrastructure on real-world geospatial layouts. In *PiperABM*, infrastructure includes homes, markets, and the roads connecting them. The library provides lightweight primitives to enable agents with a customizable decision loop, infrastructure with spatial graphs whose effective travel costs evolve, and built-in analysis for accessibility to resources and travel distance, plus optional animation. Because it is implemented with familiar scientific tools like NumPy and NetworkX, researchers can inspect and extend models directly, integrate empirical data, and reuse standard workflows. *PiperABM* supports resilience research by examining how people's choices and infrastructure conditions co-evolve, seeking to reduce the setup overhead typical of one-off models while remaining flexible enough for diverse hazards, places, and policies.

Statement of Need

One established definition describes resilience as a community's capacity to withstand, adapt to, and recover from disruptions while maintaining or restoring essential functions ([United Nations Office for Disaster Risk Reduction, 2017](#)). *PiperABM* provides a flexible foundation for resilience research: users can adopt study-specific definitions and metrics, customize agent behavior and infrastructure dynamics, and analyze simulation outputs accordingly.

Agent-based models (ABMs) represent systems "from the bottom up" using heterogeneous, interacting agents whose local rules and behaviors generate macro-level patterns ([Joshua M. Epstein, 1999](#); [Joshua M. Epstein & Axtell, 1996](#); [Schelling, 2006](#)). ABM has also been highlighted as a framework that engages diverse stakeholders, enables the forecasting of equitable futures, and empowers policy ([Hicks et al., 2021](#)). Over recent decades, ABM has matured into a practical research method supported by the scientific Python ecosystem ([Hoeven et al., 2025](#)). This paradigm is well-suited to infrastructure and community resilience, particularly within the Food–Energy–Water (FEW) nexus, where access to essential resources depends on both physical networks (e.g., roads linking homes to markets) and human decisions, and where nonlinear feedbacks, thresholds, and cascades are common ([Cansino-Loeza et al., 2022](#); [Xue et al., 2024](#); [Zhang et al., 2019](#)).

Resilience studies require graded performance and progressive degradation, such as road deterioration that increases travel costs ([Altarabsheh et al., 2021](#); [Llopis-Castelló et al., 2024](#)). They also require two-way feedback between infrastructure and human behavior ([Anshuka et al., 2022](#); [Michaelis et al., 2020](#)), and geospatial layouts for place-based analysis ([Wang et al., 2022](#)). Reproducible ensemble experiments need consistent reporting and data handling for parameter sweeps, stochastic replications, and uncertainty and sensitivity analysis ([Grimm et al., 2020](#); [McCulloch et al., 2022](#); [Razavi et al., 2021](#)). Accessible model state supports

42 custom metrics, including spatiotemporal resource accessibility under disruption (Enderami et
43 al., 2024; Tariverdi et al., 2023).

44 PiperABM is built to address these needs and reduce the setup effort required for place-based
45 resilience studies. PiperABM has been used to investigate how infrastructure degradation,
46 household size, and disruption patterns affect resource accessibility and social-network
47 robustness in an Arctic community context (Noorghasemi, Xiao, & McComb, 2025;
48 Noorghasemi, Xiao, Sha, et al., 2025).

49 Intended users are researchers and practitioners in infrastructure and community resilience,
50 including urban planners, policymakers, and decision-makers who need transparent, reproducible,
51 place-based simulations.

52 Comparison to Existing Tools

53 Among general frameworks, Mesa provides agent definition, scheduling, data collection, and
54 browser visualization (Hoeven et al., 2025); modelers supply infrastructure degradation,
55 accessibility metrics, and behavior–infrastructure coupling. InfraRisk models hazards, cascades,
56 and recovery in interdependent power, water, and transport networks (Balakrishnan &
57 Cassottana, 2022), with human decision-making typically added separately. Repast4Py
58 supports distributed and parallel ABMs (Collier & Ozik, 2022), leaving resilience-specific
59 logic to modelers. PiperABM integrates continuous degradation, resource accessibility, and
60 coupled behavior–infrastructure dynamics with customizable decision rules, reducing setup
61 effort for resilience studies.

62 Purpose-built models address multilayer recovery after Hurricane Harvey (Xue et al., 2024),
63 household recovery after Sandy (RecovUS) (Moradi & Nejat, 2020), storm-induced power-
64 outage restoration with crew dispatch (Walsh et al., 2018), community resilience under tornado
65 hazards (Aghababaei & Koliou, 2023), and flood risk–insurance dynamics (Dubbelboer et
66 al., 2017). These models serve specific studies, and adapting them to other settings often
67 requires reimplementing similar components. PiperABM provides reusable components and
68 extension points for adapting models to new hazards, geographies, and policies, while supporting
69 consistent scenario comparisons and analysis with standard Python tools.

70 Model Overview

71 PiperABM couples infrastructure and society networks represented as NetworkX graphs, with
72 their interactions summarized in Figure 1.

73 The infrastructure network represents the physical layout: nodes represent homes, markets,
74 and junctions; edges represent streets and access connections from homes and markets to
75 streets. Spatial layouts can be constructed manually or imported from map data.

76 Society nodes represent individual agents, with family, neighbor, and friendship ties as edges.
77 Family ties connect agents assigned to the same home. Agents have a home assignment,
78 current location, monetary balance, socioeconomic status, resource inventories, and sufficiency
79 thresholds, with food, water, and energy as the default resources. Home assignments and
80 agent locations link the two networks. Markets hold resource inventories, while social ties
81 identify friends' and neighbors' homes as potential destinations.

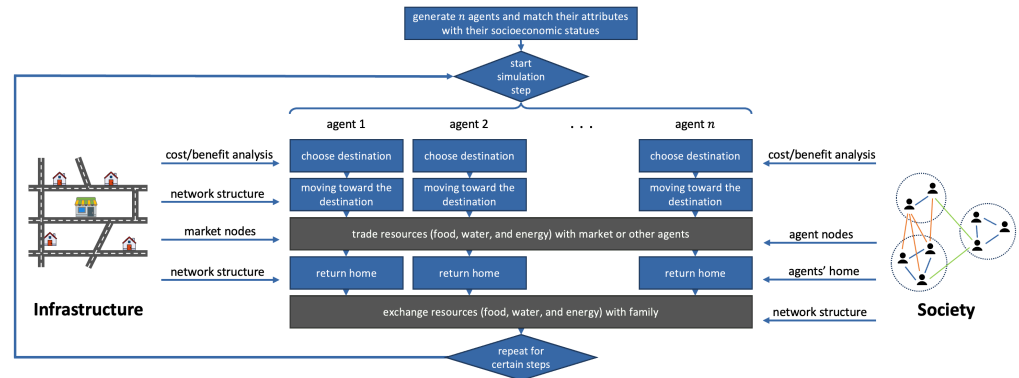


Figure 1: PiperABM overview: Infrastructure locations and social relationships shape agents' travel and resource exchanges.

The baseline decision mechanism follows OODA-style logic: observing, orienting, deciding, and acting (Brehmer, 2005; Johnson, 2023). When an agent's queued actions finish, it selects a new destination, considering markets first and socially connected homes if no suitable market is found. Destinations are ranked by resource value minus estimated travel resource costs, with travel-time constraints determining feasibility. Agents queue travel, a stay at the destination, a return home, and rest. At each simulation step, the model coordinates resource exchanges among agents at the same home or with markets where agents are present, using a cooperative, centralized Nash bargaining formulation (Nash, 1950). Market inventories are replenished each step.

Agents consume resources while staying and traveling, and travel accumulates usage on infrastructure edges. With degradation enabled, usage and aging increase streets' effective lengths, which determine A* routing costs (Foead et al., 2021) and travel duration. Infrastructure conditions therefore affect subsequent routes and feasible destinations, while agent movement contributes to further degradation.

Outputs include travel distance and resource accessibility, measured as inventory relative to a sufficiency threshold and capped at one. KeepDelta can store human-readable changes in model state (Noorghasemi & McComb, 2025), and optional animation supports face-validity assessment. Researchers can replace the baseline decision and degradation rules through user-defined extensions.

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